arXiv:2605.17789v1 [cs.CL] 18 May 2026

SocialMemBench: Are AI Memory Systems Ready for Social Group Settings?

Olukunle Owolabi
Independent Researcher
olukunle.owolabi@alumni.tufts.edu

Abstract

Memory systems for AI assistants were built for single-user dialogue and fail characteristically when applied to multi-party social group settings. This gap matters for the social assistants being built today: group-acting agents embedded in chat platforms, and proactive personal-assistant agents whose holistic model of a user must include their social context. Existing memory benchmarks evaluate dyadic or workplace dialogue; none targets multi-party social groups, where memory must anchor facts in shared history rather than professional roles, separate group norms from individual exceptions, and correctly attribute even after member departure.

We introduce SocialMemBench,[1] a benchmark of human-verified synthetic social group networks across five archetypes and three group-size tiers (4–30 members), with 430 personas and 7,355 conversation turns, yielding 1,031 QA pairs across nine question categories. Each category isolates an architectural capability, and five failure modes are formulated as testable hypotheses; our two research probes Subject-Mem and SMG provide evidence on two, three remain open.

A full-context Gemini 2.5 Flash reference reaches only 0.721 against a blind-critic reasoning-model mean of 0.98 on small networks, indicating the benchmark is genuinely difficult even with complete access to the conversation. Across all 43 networks, the four open-source memory frameworks evaluated (Mem0, LangMem, Graphiti, Cognee) cluster in the 0.12–0.18 question-weighted range with overlapping 95% CIs, well below an uncompressed retrieval reference of 0.345 and a matched-answerer full-context reference of 0.369 (GPT-4o-mini). Subject-Mem raises attribution accuracy from the 0.25–0.46 framework range to 0.78; SMG raises group-decision accuracy from 0.56–0.61 to 0.69. Closing this gap is a precondition for deployment in social group settings.

1 Introduction

AI assistants are increasingly deployed alongside multi-person conversations in two distinct modes. Group-acting agents embedded in social and messaging platforms take actions on behalf of the group;[2] proactive personal-assistant agents (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023) build holistic models of their users, for which social context is a vital input. Both classes share a single dependency: a memory layer that retains social facts across sessions and attributes them to the right person.

Current memory frameworks were designed for a single user (Zhang et al., 2025). Mem0 (Chhikara et al., 2025) keys facts to a single user; LangMem (LangChain AI, 2024) namespaces memory under that user; Graphiti (Rasmussen et al., 2025) merges same-named entities into canonical nodes; HippoRAG (Gutiérrez et al., 2024) and Generative Agents (Park et al., 2023) both treat memory as a single stream owned by one observer. These designs are correct for the dyadic chatbot setting they were built for, but break in social groups, where attribution must rest on shared history and informal norms rather than the professional roles that partially scaffold workplace dialogue. The architectural problems are concrete: a statement one member makes about another is stored against the speaker, not the subject; group consensus overwrites individual dissent rather than preserving it as an exception; a member's preferences are versioned by recency rather than by who held what

[1] Dataset: https://huggingface.co/datasets/anon4data/socialmembench (CC BY 4.0). Pipeline and audit viewer: https://anonymous.4open.science/r/SocialMemBench/ (MIT).

[2] Examples include Meta AI in WhatsApp group chats, Salesforce Agentforce in Slack channels, and Apple Intelligence's on-device personal-context retrieval (Gunter et al., 2024).

when. Speaker-aware reasoning failures in multi-party dialogue are documented at the LLM level (Kim et al., 2023b; Sclar et al., 2023), and attribution remains a challenge in workplace conversations even with full-context oracles (Hu et al., 2026). The social setting, where role-scaffolding is absent and attribution must come from the structure of the conversation itself, has not been benchmarked at the memory-architecture level.

This gap is not directly addressed by existing memory benchmarks. Prior multi-session memory benchmarks operate in dyadic chat or workplace settings: LongMemEval (Wu et al., 2025) evaluates user-to-assistant memory across five ability dimensions but is dyadic, and EverMemBench (Hu et al., 2026) tests multi-party attribution in a workplace setting where professional roles partially scaffold attribution. Social group dialogue has no such scaffolding; memory must anchor facts in shared history rather than professional roles, separate group norms from individual exceptions, and correctly attribute even after a member leaves the group. We introduce SocialMemBench to fill this cell: human-verified synthetic social group networks across five archetypes (close friends, family, recreational, interest community, acquaintance network) and three group-size tiers (4–30 members), with 430 personas, 348 sessions, 7,355 conversation turns, and 1,031 QA pairs across nine social-memory question categories, evaluated session-by-session against four open-source memory frameworks, two research probes, and full-context baselines.

The goal is not only to measure performance but to explain it. For each system we examine per-category results alongside the design choices its adapter implements, identify recurring patterns as five failure modes, and use targeted probes to test whether specific structural changes improve performance on the affected categories. The findings are directly actionable for any team building memory for a group-deployed AI assistant.

The paper's contributions are: a corpus of human-verified synthetic social group networks across five archetypes (close friends, family, recreational, interest community, acquaintance network), three group-size tiers (4–30 members), and 43 networks total, with 430 personas, 348 sessions, 7,355 conversation turns, and 1,031 QA pairs across nine question categories; an evaluation of four open-source memory frameworks and our research probes Subject-Mem and SMG, with evidence on two of five failure modes and three open; a taxonomy of five observed failure patterns grounded in the results; and an open evaluation pipeline with a browser-based data audit viewer used for human review during dataset construction.[1]

2 Related Work

Memory and Long-Context Benchmarks. The challenge of preserving facts across extended conversations has been studied at progressively larger scale. Xu et al. (2022) introduced a canonical dataset of persona-grounded dyadic conversations spanning up to five sessions, establishing multi-session coherence as an evaluation target. Maharana et al. (2024) scaled this to 32-session dyadic dialogues with persona and temporal-event-graph grounding, averaging 600 turns per pair, and added multi-hop and temporal QA. Wu et al. (2025) formalized user-to-assistant memory as five ability dimensions (single-session extraction, cross-session reasoning, temporal reasoning, knowledge updates, and abstention), finding 30–60% accuracy drops when chat histories span multiple sessions. Hu et al. (2026) breaks the dyadic ceiling: it tests multi-party workplace conversations and reports that multi-hop attribution accuracy reaches approximately 26% even with oracle model access to the full conversation. A separate body of work evaluates whether LLMs can reason over long documents within a single context window. SCROLLS (Shaham et al., 2022) established a unified suite of long-context tasks spanning summarization, question answering, and natural language inference, and subsequent benchmarks have progressively extended context lengths while showing that claimed context windows overstate real capability.

Memory Frameworks. Several open-source memory frameworks for LLM agents have emerged. Mem0 (Chhikara et al., 2025) extracts atomic facts with semantic deduplication; Graphiti (Rasmussen et al., 2025) builds a temporally-aware entity-relationship graph with validity windows and provenance; LangMem (LangChain AI, 2024) uses namespace-keyed storage with reflection-driven consolidation; Cognee (topoteretes, 2024) combines graph and vector storage with LLM-extracted facts. MemGPT (Packer et al., 2023) introduced tiered virtual memory with LLM-driven paging, and Generative Agents (Park et al., 2023) layered recency, importance, and relevance scoring over a natural-language memory stream. Hip-

poRAG (Gutiérrez et al., 2024) applies graph-structured retrieval with personalized PageRank over LLM-extracted entities.

Personalization and Social Structure. Personalization research addresses individual user modeling. LaMP (Salemi et al., 2024) established the canonical benchmark for single-user personalization, and subsequent retrieval- and parameter-based work has used it as the evaluation target. PersonaChat (Zhang et al., 2018) and ConvAI2 model persona as short in-context text fragments rather than maintained cross-session state. Social NLP has studied structure within conversations. DialogRE (Yu et al., 2020) extracted typed relations between speakers within single dialogues; graph-based emotion recognition models encode speaker structure per session. Gandhi et al. (2023) provide a broad survey of social reasoning capabilities in language models, finding that even large models fail to reliably track belief states across multi-agent interactions. FANToM (Kim et al., 2023b) tests theory-of-mind reasoning in multi-party conversations where information asymmetry arises from characters with unequal access to prior exchanges.

Table 1 compares these benchmarks across architectural dimensions.

3 Benchmark Design

3.1 Task Definition

Let $G = (P, E, S)$ be a social group with personas P, pairwise relationship edges E, and a session sequence $S = (s_1, \ldots, s_T)$. A memory system M ingests sessions sequentially, extracting facts and constructing a memory state after each one. After all sessions are ingested, M answers a question q about a specific persona or group interaction using only that memory state; the original conversation is not re-supplied as context. SocialMemBench measures how accurately M can answer q under this setup. Upper-bound references are reported separately by giving an LLM answerer the full conversation as context rather than a memory state.

3.2 Query Taxonomy

The benchmark covers nine proposed question categories, each targeting a distinct capability that a social-group AI must have. Table 2 lists the categories with examples and the key difficulty driver for each. Each category contains many questions across many networks; the labels are category identifiers, not question numbers.

Three categories test attribution specifically: attribution probes (AP, direct speaker confusion), theory-of-mind (TM, who knows what about whom), and norm-vs-individual (NI, does a group-level fact apply to this specific person).

3.3 Data Generation Pipeline

Producing realistic group conversations with verifiable ground truth requires three stages. Full construction details and acceptance criteria are in Appendix A.

The corpus is synthetically generated. Synthetic generation is not a shortcut but a methodological choice: it lets us plant ground-truth challenges at known positions, enforce per-persona voice fingerprints verified by a blind critic, and control implicitness levels across nine question categories. BABILong (Kuratov et al., 2024) embeds synthetic needles in long-context distractors; SODA (Kim et al., 2023a) distills synthetic socially-grounded dialogues from LLMs; RULER (Hsieh et al., 2024) uses synthetic construction to control evaluation properties precisely; Self-Instruct (Wang et al., 2023) bootstraps synthetic instruction data. These benchmarks use generation for the same reason: to guarantee evaluation properties that naturally-occurring data cannot provide.

Stage 1: Persona Network. Each network specifies n personas with personality profiles, communication styles, preference histories, and pairwise relationship edges. Structural constraints are enforced at generation time and verified before proceeding: at least one contested group norm (ground truth for NI questions); at least one preference history event per persona (ground truth for TS questions); distinct communication profiles across personas. Networks span five group types (close friends, family, recreational, interest community, acquaintance network) and three size tiers (Table 3).

Stage 2: Conversation Corpus. Each session captures a discrete period of group conversation: a self-contained WhatsApp-style chat thread with multiple speakers contributing turns. For each network we generate S sessions, rendered with session and speaker headers as ingested by the memory systems. Each session embeds 2–4 planted challenges: ground-truth facts expressed implicitly through behavior, complaint, silence, or deflection rather than direct statement. Each challenge records the subject persona, the speaker, the fact, and a verba-

Table 1: Benchmark comparison across architectural dimensions. "Multi-party" = ≥3 co-present speakers per session. "External memory" = tests a write-index-retrieve layer across discrete sessions where the original conversation is not re-supplied at query time (vs. LLM reasoning over full context). "Social setting" = informal social group (friends, family, recreational, interest community), not workplace, dyadic strangers, or single-user-to-assistant. "Member departure" = corpus contains members who leave mid-corpus with explicit ground truth for pre-departure state recall.

Benchmark	Multi-party	External memory	Social setting	Member departure
MSC (Xu et al., 2022)	–	–	–	–
LoCoMo (Maharana et al., 2024)	–	–	–	–
LongMemEval (Wu et al., 2025)	–	✓	–	–
LaMP (Salemi et al., 2024)	–	–	–	–
FANToM (Kim et al., 2023b)	✓	–	–	–
EverMemBench (Hu et al., 2026)	✓	✓	–	–
SocialMemBench (ours)	✓	✓	✓	✓

Table 2: SocialMemBench query categories. Each targets a distinct memory capability. Category codes are used as compact column headers in results tables. Inline tags show the archetype the example came from and the answer format: "OE" = open-ended (LLM-judge scored 0–1), "MC" = multiple-choice (scored by exact letter match).

	Category	Example question	Difficulty driver
SR	Single-contact recall	What does Hiro's behavior in reunion food planning suggest about his eating pattern, and how is it visible without him stating it directly? (close friends · OE)	Implicit signal; multi-turn synthesis
GD	Group decision recall	When Bex closes the headcount for the rooftop component, what did Aanya's response signal about her actual position? (close friends · MC)	Consensus decision plus dissenter identity
MA	Multi-contact aggregation	All five Maple Street members shared their views on the residents' parking scheme. What was each person's position and reasoning? (acquaintance · OE)	Per-persona enumeration; some preferences implicit
AP	Attribution probe	Who expressed the view that the value of a photography session lies in intentionality rather than the number of exposures? (interest community · MC)	Two plausible speakers; attribution by meaning
TM	Theory of mind	What is Priya's concern about running on uneven surfaces, and what in Fatima's response shows she already knew about this? (recreational · OE)	Cross-persona knowledge revealed by behavior
NI	Norm vs. individual	The group has a norm of raising issues in the chat before contacting the council. Does that norm apply to everyone? (acquaintance · MC)	Norm-exception distinction; implicit dissent
RE	Relational edge	What does the conversation reveal about Bram and Lina's relationship, particularly through the First Watt loan exchange? (interest community · OE)	History embedded in passing references
TS	Temporal shift	What in the conversation signals that Priya's relationship to running and team sport has shifted, and what appeared to drive it? (close friends · OE)	Old state + trigger + new state; cross-session
DM	Departed member	Based on what the group knew about Bex before she left, what was her relationship to cycling? (close friends · OE)	Stale-state recall post-departure

tim evidence span in the message that expresses it. Challenge types cover the query taxonomy (see Table 6 in the appendix).

Stage 3: QA Generation and Verification. Each planted challenge seeds one or more questions. A pre-generation inference gate rejects any question answerable by keyword search and enforces that difficulty labels match required reasoning depth. After generation, a blind-critic pass (an LLM evaluator shown only the plain-text corpus, with no access to planted challenge metadata) independently answers every question and scores it against the ground truth. A network is accepted only when the blind-critic mean score ≥ 0.70 and all evidence spans verify verbatim. Every QA pair was additionally reviewed by a human using the browser-based data audit viewer (Appendix F), checking plausibility, attribution correctness, and answer completeness; pairs failing this review were revised or removed (full criteria in Appendix A).

3.4 Dataset

The corpus is 348 sessions of natural group dialogue across 43 networks (430 personas, 660 pairwise relationship edges): 7,355 conversation turns

averaging 21.1 turns per session, 292,725 tokens total under the GPT-4o-mini tokenizer. Each session is a self-contained chat exchange generated to embed 2–4 planted challenges; each challenge records the subject persona, the speaker, a verbatim evidence span, and an implicitness level. Of the 1,091 planted challenges, 89% carry implicitness level 2 or 3, meaning the fact is expressed through behavior, complaint, or deflection rather than stated directly. Every question in the benchmark has a traceable ground-truth anchor in the text: a specific turn where the relevant fact is revealed. Networks span five group types (close friends, family, recreational, interest community, acquaintance network), covering the principal social contexts in which group-deployed AI assistants operate. Attribution patterns, norm structures, and relationship histories differ meaningfully across group types; a system that handles only close-friend networks has not solved the general problem.

Questions cover all nine categories listed in Table 2. Blind-critic mean scores range 0.93–0.98 across categories (overall mean 0.95), confirming every question is answerable from the corpus by a reader with full access. A system that stored only explicitly stated preferences would score near zero on the 89% of questions backed by implicit challenges.

4 Evaluation Framework

4.1 Design

We evaluate four open-source memory frameworks representative of current architectural approaches: GRAPHITI (Zep AI), LANGMEM, MEM0, and COGNEE. We also evaluate two research probes built for this benchmark, each a minimal architectural variant of a standard framework. SUBJECTMEM (Subject-Indexed Memory) indexes facts by the subject persona rather than the speaker. SMG (Social Memory Graph) uses a structured graph with dissent-aware and relational edges instead of flat embeddings.

Two reference baselines hold the answerer constant for isolation of memory-layer effects. The *full-context baseline* (LLM-MINI, GPT-4o-mini) receives the complete corpus as input context with no memory system or retrieval step. The *uncompressed retrieval baseline* stores every conversation turn in a vector store without extraction or summarization, retrieving by cosine similarity at query time; it is not a memory system and could not operate at scale, but it shows what raw retrieval recovers without a memory layer.

All conditions use GPT-4o-mini as the answerer and top-k=10 retrieval where retrieval is used. The non-reasoning answerer is a deliberate design choice: a strong reasoning model with extensive chain-of-thought can compensate for incomplete or misattributed retrievals through inference, masking the architectural failure modes the benchmark is built to expose. The architectural claims we make in Sections 5–6 describe information loss prior to retrieval and are model-agnostic in this sense; absolute score levels are configuration-specific. Each framework runs under this matched configuration rather than its out-of-the-box default, so that performance differences trace to architectural choices in storage and indexing rather than configuration disparities. Implementation details are in Appendix B.

4.2 Evaluation

Session-by-session evaluation reflects how memory systems operate in deployment: each session is ingested once, the system extracts facts and updates its memory state, and queries are answered using only that retained state without the corpus being re-supplied. This is meaningfully harder than full-context inference, and it is the condition under which any group-deployed AI assistant must operate. The setting is especially relevant for smaller production models, where accurate long-term memory and careful context management are required to compensate for limited reasoning. Sessions are ingested in chronological order; all queries are issued after the final session.

Multiple-choice categories are scored by exact letter match against the correct option. Open-ended categories are scored 0–1 by an LLM judge (GPT-4o-mini) calibrated to avoid verbosity bias: a terse correct answer scores the same as a detailed one. Full judge rubric and per-category scoring rules are in Appendix C.

Results cover all 43 networks across all three tiers: 19 small-tier (264 questions, including 4 networks with member-departure scenarios), 14 medium-tier (346 questions), and 10 large-tier (421 questions, networks ranging 15–30 members). All conditions were evaluated on the same sets of networks at each tier.

Table 3: Corpus statistics by network-size tier. All 43 networks passed structural verification, blind-critic QC, and were evaluated across all 8 conditions (19 small including 4 with member-departure scenarios, 14 medium, 10 large).

Tier	Nets	Members	Personas	Sessions	Turns	QA
Small	19	4–6	95	122	2,183	264
Medium	14	8–10	130	124	2,562	346
Large	10	15–30	205	102	2,610	421
Total	**43**		**430**	**348**	**7,355**	**1,031**

5 Results

5.1 Data Quality and Reference Bounds

Before comparing memory systems, we establish three reference points: whether the answers are recoverable from the conversations, how the matched answerer performs when the memory layer is bypassed, and how raw-turn retrieval performs when no extracted memory state is built. The blind critic, Claude Sonnet 4.5, serves as a validation reference for answer recoverability: it has full conversation access and no planted-challenge metadata, and independently re-answers each question. It reaches a mean score of 0.952. Human reviewers also inspected the released QA pairs in the browser-based audit viewer for evidence grounding, attribution correctness, and answer completeness. Together, these checks show that the gold answers are recoverable from the conversations and supported by the cited evidence.

The two all-network performance references isolate the answerer and retrieval components from memory architecture. The matched full-context GPT-4o-mini reference reaches 0.369, showing how the answerer performs when the memory layer is bypassed entirely. The uncompressed raw-turn retrieval reference, also answered by GPT-4o-mini, reaches 0.345, showing how retrieval performs when every conversation turn is indexed directly and no extracted or consolidated memory state is built. Neither condition is a memory-system competitor; they provide reference points for the memory-system results below.

5.2 Memory System Performance

Table 4 and Figure 1 show scores across all 43 evaluated networks.

Three memory-system results stand out. First, all four memory frameworks cluster in the 0.12–0.18 question-weighted mean range, well below the uncompressed retrieval reference (0.345). Bootstrap 95% confidence intervals (10,000 resamples over networks) are shown alongside network-weighted means in Table 4. Pairwise bootstrap tests show that no two open-source memory frameworks are statistically distinguishable on aggregate. Second, Subject-Mem significantly outperforms the strongest framework, Graphiti ($p < 0.001$). SMG is higher than Graphiti in mean score but not significantly so on aggregate ($p = 0.123$); its main gains appear in specific categories, especially group decisions and norm exceptions. Third, departed-member retention separates the systems most starkly. Graphiti scores 0.00 on Q9, while Mem0 (0.10) and Cognee (0.07) remain near floor. Subject-Mem (0.35), SMG (0.23), and the uncompressed retrieval reference (0.34) retain more departed-member signal, though none solves the category. Section 6 analyzes the architectural mechanisms behind this pattern.

5.3 Performance by Category

Aggregate scores hide category-level variation. Figure 3 shows that conditions separate differently by category: attribution probes favor conditions that preserve speaker and subject information, group-decision and norm-exception categories show the clearest gains from the social-graph probe, and theory-of-mind and temporal-shift categories remain low across all evaluated memory frameworks.

Attribution probes show the widest spread. The uncompressed retrieval reference scores 0.82 on AP, and Subject-Mem reaches 0.78; the four memory frameworks span 0.25–0.46. Both high-AP conditions preserve speaker information at ingestion (raw turn text and subject-indexed extraction, respectively); the frameworks compress speaker information at ingestion and lose it. The 0.32-point gain from Subject-Mem over the best framework (0.78 vs. Graphiti 0.46) is achieved with the same answerer, retriever, and corpus, isolating ingestion-time indexing as the source of the gain.

SMG's category-level gains isolate the contribution of dissent-aware graph structure. Holding

Table 4: Mean scores by condition and question category across all 43 evaluated networks. All conditions use GPT-4o-mini as the answerer for matched comparison: LLM-MINI is the full-context matched-answerer reference, and UNCOMPRESSED RETRIEVAL indexes raw conversation turns without extracted memory state. Mean^Q is question-weighted; Mean^N is network-weighted with bootstrap 95% confidence intervals from 10,000 resamples over networks. Category codes match Table 2.

Condition	Mean^Q	Mean$^N \pm$ 95% CI	SR	GD	MA	AP	TM	NI	RE	TS	DM
Reference conditions (held constant across systems)											
LLM-MINI (full context)	0.369	0.406 ± 0.054	0.34	0.65	0.12	0.75	0.31	0.59	0.23	0.28	0.38
UNCOMPRESSED RETRIEVAL	0.345	0.357 ± 0.028	0.34	0.63	0.12	0.82	0.21	0.44	0.24	0.24	0.34
Memory frameworks (open-source, current architectural approaches)											
GRAPHITI	0.178	0.201 ± 0.043	0.11	0.61	0.06	0.46	0.05	0.41	0.12	0.07	0.00
LANGMEM	0.162	0.177 ± 0.033	0.08	0.56	0.15	0.39	0.06	0.44	0.06	0.09	0.15
MEM0	0.143	0.155 ± 0.032	0.11	0.57	0.05	0.31	0.05	0.24	0.06	0.07	0.10
COGNEE	0.120	0.125 ± 0.025	0.09	0.59	0.02	0.25	0.10	0.20	0.05	0.01	0.07
Research probes (social-aware indexing ablations)											
SUBJECT-MEM	0.321	0.318 ± 0.035	0.31	0.61	0.06	0.78	0.24	0.46	0.19	0.22	0.35
SMG	0.213	0.226 ± 0.031	0.16	0.69	0.08	0.44	0.11	0.46	0.12	0.10	0.23

the answerer constant, SMG scores 0.69 on GD against the framework range of 0.56–0.61, and 0.46 on NI, matching the best alternatives (LangMem 0.44, uncompressed 0.44). The gain is recoverable because dissent edges retain information that flat-embedding stores discard at ingestion. Aggregate SMG (0.213) sits below the uncompressed reference (0.345); the result is targeted recovery on graph-structured categories, not overall dominance.

TM and TS are the lowest categories across all conditions, including the matched-answerer reference (TM 0.31, TS 0.28). The floor has two compounding causes: the deliberate non-reasoning answerer (Section 4.1) and the architectural gaps shared by every evaluated framework. Section 6 catalogues the gaps. A matched-oracle comparison against Gemini 2.5 Flash confirms the answerer is not the architectural bottleneck (Appendix D).

6 Failure Mode Analysis

The performance gaps in Table 4 trace to specific architectural decisions in each framework. This section pairs each failure mode with the design choice that produces it and the schema primitive that would address it; Table 5 summarises all five modes.

Attribution failures have two architectural causes. The first is single-stream conflation: frameworks that ingest all turns into a shared store (Cognee, LangMem, Mem0) flatten "Sarah said X about Mike" into a fact about Sarah, or into a group-level summary. Subject-Mem changes only the indexing layer, extracting (subject, fact) pairs and storing facts under the persona they describe; AP rises from the framework range of 0.25–0.46 to 0.78 (Table 4). The fix is per-persona attribution at ingestion.

The second cause is missing cross-persona representation. Theory-of-mind questions require representing what A revealed about B's preference, not just what B prefers. No evaluated system stores this; TM scores cluster at 0.05–0.10 across the four frameworks. The fix is a cross-persona edge primitive (`KNOWS_ABOUT`), absent from every framework we evaluated.

Temporal-state failures trace to overwrite-on-update: when a preference changes, the evaluated adapters update the current value without preserving the prior state. Questions of the form "what drove the change, and what was the prior state?" cannot be answered from a single current value. Graphiti exposes validity intervals via its bi-temporal model in principle, but the default extraction pipeline does not emit prior-state edges when a new fact arrives. SMG implements an `EVOLVED_FROM` edge that links each new fact to the prior state it replaced; SMG still scores 0.10 on TS, and the matched-answerer reference reaches only 0.28, so representation alone does not close the gap. The storage-layer fix is `EVOLVED_FROM` plus an extraction step that emits it when a fact contradicts a prior.

Entity merging is the failure mode that most clearly traces to a single framework decision. Graphiti's ingestion pipeline canonicalises personas into shared graph nodes; the large-tier collapse on SR, NI, RE, and TS at 15+ members traces to merging cascades, not to per-network configuration. The clearest signature is departed-member retention: Graphiti scores 0.000 on Q9 across all

4 evaluated DM networks because subsequent ingestion overwrites the merged-out persona's facts. Subject-Mem and the uncompressed retrieval reference handle Q9 (0.35 and 0.34 respectively) because neither merges or summarises persona state at ingestion.

Norm-individual conflation surfaces when a group norm has an exception that requires per-persona representation. SMG's dissent-aware schema raises GD to 0.69 against the framework range of 0.56–0.61 and NI to 0.46 against LangMem's 0.44, locating the fix at the schema layer.

Our research probes target two of the five modes (Subject-Mem on AP; SMG on GD/NI); the other three remain open.

Performance follows a U-shape across tiers (Figure 2): small-tier corpora concentrate signal per persona, medium-tier sits in a difficult middle, and large-tier recovers as more sessions provide supporting attribution evidence. The pattern is most pronounced for full-context models (LLM-MINI: 0.476 small, 0.257 medium, 0.395 large). Question-type mix is nearly identical across tiers (AP 7–10%, TS 20–27%, SR 19–23%), so the dip is not a selection artifact.

Memory frameworks show the same shape at lower amplitude (0.12–0.22 small, 0.13–0.16 medium, 0.10–0.17 large; Cognee declines monotonically to 0.10). The plateau is consistent with a fixed retrieval budget: top-k chunks from a growing store return a lower fraction of relevant context per query, and the probability that a fact is stored under the wrong persona increases as more speakers share the same index. Graphiti's large-tier profile shows TS at 0.078 floor with across-network variance on NI (0.263) and RE (0.124), consistent with merging cascades that depend on persona similarity rather than group size alone.

7 Conclusion

SocialMemBench exposes a consistent gap between what current memory frameworks achieve and what social group settings require. Across 43 networks, the four evaluated frameworks cluster at 0.12–0.18, well below the uncompressed retrieval reference (0.345) and the matched-answerer full-context reference (0.369). The gaps concentrate on attribution, theory of mind, temporal evolution, norm exceptions, and departed-member retention. Each of these maps to a capability any AI assistant deployed in a group chat needs to handle: tracking who said what about whom, modelling what one persona has learned about another, recognising when a preference has changed, distinguishing group norms from individual exceptions, and retaining facts about members who leave. The benchmark makes social-memory architecture a tractable architectural research problem: each of the five failure patterns is a testable architectural hypothesis, two evidenced in this work by Subject-Mem and SMG, and three that define a research agenda.

Closing the full gap requires combining: per-persona attribution slots; temporally-ordered preference histories; entity-lifecycle preservation across membership changes; cross-persona knowledge edges; and dissent-aware group-norm schemas. Two are reachable as adapter-layer changes today (per-persona attribution, dissent-aware schema); one requires extraction-pipeline and storage-schema work (temporal histories); two require framework-level changes (entity-lifecycle preservation, cross-persona knowledge). Researchers can use SocialMemBench to develop fixes for the named failure patterns and to study social memory architectures more broadly. Closing this gap is a precondition for deployment of memory systems in social group settings.[1]

Limitations

We see several directions for future work. The synthetic generation pipeline trades register variation and naturalism for controlled experimental variables: planted-challenge ground truth at known (speaker, session, turn) coordinates, controlled distribution of rare attribution patterns (theory-of-mind reveals, member-departure dynamics with retained-state probes), and verifiable evidence anchors. Real-data extension is a natural complement, capturing what synthetic generation cannot while requiring different methodological tools to handle the absence of ground truth. Our pipeline models per-persona communication profiles, implicit preferences, relationship histories, and group norms (the structural elements our failure modes target).

The evaluation methodology reflects deliberate tradeoffs. The QA quality check shares a model family with the data generator (Claude Sonnet 4.5 and Claude Code); per-pair human review via the audit viewer (Appendix F) and downstream evaluation across different model families (GPT-4o-mini in Section 5, Gemini 2.5 Flash in Section D) bound the family-overlap risk, with independent annota-

Table 5: Five failure patterns observed in evaluated adapter configurations, the systems affected, and the categories where they surface. Claims are specific to the adapter versions and configurations tested. No evaluated system addresses all five. Categories give the measurement target for each failure mode.

Failure pattern	**Affected adapters**	**Categories**	**Fix requires**
Single-stream conflation	Cognee, LangMem, Mem0	SR, AP	Per-persona attribution slots
Temporal state overwrite	Graphiti, SMG	TS	Ordered fact history (`EVOLVED_FROM`)
Entity merging at scale	Graphiti	DM, large-tier NI/RE/TS	Entity-lifecycle preservation
No cross-persona knowledge	All systems	TM	Cross-persona edges (`KNOWS_ABOUT`)
Norm-individual conflation	Cognee, LangMem, Mem0, Graphiti	NI, GD	Dissent-aware schema

tion a natural extension. Memory-system evaluations use a single answering model (Section 4.1) to isolate memory-layer effects from reasoning capacity; the small-tier oracle comparison (Section D) shows architectural failure modes are answerer-independent, and extending to stronger answerers would need to control for chain-of-thought masking the architectural failures we measure. Open-ended scoring uses a single GPT-4o-mini judge calibrated against verbosity bias, attribution error, and non-answers (Zheng et al., 2023); multi-judge or human evaluation would tighten the scoring claim at the cost of inter-rater variance on interpretive categories, and multiple-choice questions (21% of the dataset) are scored by exact string matching. The benchmark covers English-only text conversation with group sizes 4–30, holding language and modality constant to isolate architectural effects. The architectural failure modes apply at any scale where the underlying framework designs are deployed; multilingual and multimodal (memes, images, voice) extensions are natural follow-ups.

References

Prateek Chhikara and 1 others. 2025. Mem0: Building production-ready AI agents with scalable long-term memory. In *Proceedings of the 27th European Conference on Artificial Intelligence*.

Kanishk Gandhi, Jan-Philipp Stabile, Patrick Smyth, and Noah D. Goodman. 2023. Understanding social reasoning in language models with language models. In *Advances in Neural Information Processing Systems*, volume 36.

Tom Gunter and 1 others. 2024. Apple intelligence foundation language models.

Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Ryo Kamoi, and Yu Su. 2024. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*.

Yiming Hu and 1 others. 2026. Evaluating long-horizon memory for multi-party collaborative dialogues. arXiv preprint arXiv:2602.01313. ArXiv preprint.

Hyunwoo Kim, Jack Hessel, Liwei Jiang, Peter West, Ximing Lu, Youngjae Yu, Pei Zhou, Ronan Le Bras, Malihe Alikhani, Gunhee Kim, Maarten Sap, and Yejin Choi. 2023a. SODA: Million-scale dialogue distillation with social commonsense contextualization. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

Hyunwoo Kim, Melanie Sclar, Xavier Tang, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023b. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Yuri Kuratov, Aydar Bulatov, Petr Anokhin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. 2024. BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack. In *Advances in Neural Information Processing Systems*, volume 37.

LangChain AI. 2024. LangMem: Long-term memory for LLM applications. Software. `https://github.com/langchain-ai/langmem`.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. LoCoMo: Long context modeling benchmark for LLMs via naturally evolving dialogues. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. 2023.

MemGPT: Towards LLMs as operating systems. arXiv preprint arXiv:2310.08560.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.

Preston Rasmussen and 1 others. 2025. Graphiti: A temporally-aware knowledge graph architecture for complex agent workloads. arXiv preprint arXiv:2501.13956. Zep AI open-source project.

Alireza Salemi, Sheshera Mysore, Michael Bendersky, and Hamed Zamani. 2024. LaMP: When large language models meet personalization. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Melanie Sclar, Sachin Kumar, Peter West, Alane Suhr, Yejin Choi, and Yulia Tsvetkov. 2023. Minding language models' (lack of) theory of mind: A plug-and-play multi-character belief tracker. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.

Uri Shaham, Elad Segal, Maor Ivgi, Avia Efrat, Jonathan Berant, and Omer Levy. 2022. SCROLLS: Standardized comparison over long language sequences. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.

topoteretes. 2024. Cognee: Knowledge graph memory for AI agents. Software. https://github.com/topoteretes/cognee.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*.

Di Wu, Hongwei He, Wenhao Shi, Weiwei Wang, and Junwei Han. 2025. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *The Thirteenth International Conference on Learning Representations*.

Jing Xu, Arthur Szlam, and Jason Weston. 2022. Beyond goldfish memory: Long-term open-domain conversation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.

Dian Yu, Kai Sun, Claire Cardie, and Dong Yu. 2020. DialogRE: Human-annotated dialogue-based relation extraction dataset. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. 2018. Personalizing dialogue agents: I have a dog, do you have pets too? In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*.

Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025. A survey on the memory mechanism of large language model based agents. *ACM Transactions on Information Systems*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

A Dataset Construction Details

A.1 Network Generation: Schema and Constraints

Each network is stored as a JSON file (`data/raw/{network_id}.json`) with four top-level structures. The `personas` field is a list of n persona objects, each carrying display name, age, occupation, background, Big Five personality scores on a three-point scale, a seven-dimension communication profile (formality, verbosity, emoji use, humor, directness, typo rate, and warmth), categorized preferences, a preference history recording old value, new value, trigger, and session index per event, and two to three distinctive speaking quirks. The `edges` field records pairwise relationship edges with relationship type, closeness, sentiment, a history summary, and shared events. The `group_norms` field lists group-level behavioral norms, each with a boolean `truly_universal` flag and a dissenters list that serves as the Q6 ground truth anchor. The `metadata` field records sessions planned, turns per session, and departure events.

The following structural constraints are enforced at Stage 1 and verified before any network proceeds to Stage 2. Closeness ratings are not uniformly high: at least one relationship edge must carry mixed or tense sentiment. At least one group norm must be non-universal with a non-empty dissenters list. Every persona must have at least one preference history event. No two personas may share the same formality-emoji-humor triple, and every persona pair must differ on at least two of verbosity, emoji use, and formality. Edge coverage is enforced by group size: five-person networks

require at least six edges; nine-person networks require at least twelve.

A.2 Conversation Generation: Planted Challenge Types

Seven planted challenge types are distributed across sessions during Stage 2 generation; an eighth, member departure, is encoded at the network level rather than as a planted conversational signal (Table 6). Each planted challenge records: `challenge_type`, `subject_persona_id`, `speaker_persona_id`, `fact_attribute`, `fact_value`, `evidence_span` (verbatim substring of the message, $\leq$200 chars), `implicitness_level` (1–3), `session_index`, and `turn_index`.

Implicitness levels control difficulty. Level 1 preferences are stated directly and are used sparingly, mostly as foil setups for attribution probes. Level 2 (the default) requires inferring the preference from a complaint, reaction, or passing comment. Level 3 (the hardest) requires reading silence, deflection, or subject change as a preference signal.

A.3 Conversation Quality Acceptance Criteria

The Stage 2 QC blind-critic pass scores five conversation quality dimensions on a 0–2 scale. A network is accepted only when all five dimensions score ≥ 1 and `em_dash_free` scores exactly 2.

A.4 QA Generation: Inference Depth Rubric

The inference depth rubric is applied as a pre-generation gate. Any pair that fails is either reframed or skipped.

A.5 Query-Type Structural Flags

The Stage 3 QC pass applies rule-based structural flags per query type; flagged pairs are rewritten or removed. General flags reject pairs that are trivially answerable by keyword search or that embed the answer (a preference keyword or speaker name) in the question text. Type-specific flags enforce structural requirements: Q4 pairs must include a contamination foil with at least a two-sentence explanation naming a specific vocabulary or framing similarity between target and foil speaker; Q5 pairs must ask both what was revealed and who revealed it through observable chat behavior (not a claimed private conversation); Q8 pairs must have at least two temporal anchors in non-consecutive sessions and must ask for the trigger as well as the change.

A.6 Dataset File Format

Conversations (`data/conversations/{id}.json`): JSON with `sessions` list; each session has `session_id`, `date`, `date_label`, `topic`, and `turns`. Each turn has `turn_id`, `speaker_display_name`, `timestamp`, `message`, and `planted_challenges` (list of challenge objects). A flat `planted_challenge_index` at the corpus root allows fast lookup without traversing all sessions.

QA pairs (`data/qa/{id}.jsonl`): one JSON object per line. Key fields: `qa_id`, `query_type`, `difficulty`, `question`, `answer`, `answer_format` (long_form / multiple_choice / short_answer), `options`, `correct_option`, `evidence_anchors` (list of session index + turn ID + verbatim excerpt), `contamination_foil` (Q4 only), `temporal_anchors` (Q8 only), and QC fields: `qc_phase1_score`, `qc_phase2_grounded`, `qc_phase3_flagged`.

Manifest (`data/manifest.json`): incremental tracking of all networks; records status, session count, QA count, consumed challenge IDs, eval summary, and verify status per network.

B Memory System Adapter Details

All memory system evaluations follow a session-sequential ingestion protocol: sessions are processed in chronological order; the memory system is not shown future sessions during ingestion. Queries are issued after the final session has been ingested.

B.1 Adapter Interface

All adapters implement a common Python abstract base class (`eval/adapters/base.py`) with two async methods: `add()` ingests all sessions into the memory system, and `search()` retrieves relevant memories for a given question.

The adapter's `search()` output is passed verbatim as `{context}` in the answerer prompt. This means retrieval quality and formatting directly affect the answering model's performance.

B.2 Adapter Configurations

Table 9 summarizes each adapter's implementation, Python environment, and key design choices.

Table 6: Planted challenge types and their role in the QA ground truth. Codes match Table 2. The final row (member departure) is structural, encoded at the network level rather than as a planted conversational signal.

Challenge type	Definition	Seeds QA type(s)
`implicit_preference`	Subject's preference revealed through behavior or complaint, not stated directly	SR, MA
`consensus_trap`	Group makes a decision; one dissenter goes quiet or deflects	GD, NI
`false_attribution_seed`	Two personas make similar-sounding statements on related topics in distinct voices	AP
`theory_of_mind`	Speaker A reveals what they know about Speaker B's preference via an observable action	TM
`temporal_shift`	Persona's preference contradicts an earlier stated preference; change has a trigger	TS
`relational_disclosure`	History or nature of a relationship between two personas emerges in passing	RE
`relational_graph`	Group structure revealed: how members met, sub-clique dynamics, closeness signals	RE
`member_departure`	Member leaves the network mid-corpus; pre-departure facts become Q9 ground truth	DM

Table 7: Conversation quality dimensions assessed in the Stage 2 QC pass.

Dimension	Pass criteria	Score 0 trigger
Topic diversity	Sessions cover topics consistent with the group type's real behavior (recreational: activity-anchored; close-friends: distinct per-session thread; family: rotating domains)	Any session is a near-duplicate of another
Voice consistency	Per-persona voice fingerprints realized across 5 sub-dimensions: emoji use, typo rate, verbosity, formality, speaking quirks	Any persona's emoji or verbosity contradicts their profile
Transition naturalness	Topic shifts triggered by a character raising something; no cold jumps or announced pivots	Any unexplained topic switch
Em-dash free	Zero `--` in any `message` field (generation artifact absent in real chat)	Any em dash found
Planted turn integrity	All evidence spans are verbatim substrings of the corresponding turn message	Any span not found verbatim

Table 8: Inference depth rubric used to assign difficulty labels and filter trivially answerable pairs.

Depth	Label	Definition
0	Reject	Answerable by keyword or name search: one word in the transcript gives the answer
1	Easy	Single-turn contextual reading required; answer is stated in one place
2	Medium	Two-turn synthesis or one inference step; connecting two pieces of evidence
3	Hard	Cross-session inference or attribution ambiguity; tracking signals across $\geq$3 turns or distinguishing two plausible speakers by meaning

Table 9: Memory system adapter configurations. All adapters use GPT-4o-mini as the answering model; extraction model column refers to memory ingestion only.

Condition	Implementation	Env	Extraction	Retrieval
UNCOMPRESSED RETRIEVAL	Chroma vector store, raw turns verbatim	smb310	None (raw text)	Cosine similarity; `text-embedding-3-small`; top-10
SUBJECT-MEM	Chroma; indexes facts by subject, not speaker	smb310	GPT-4o-mini	Cosine similarity; `text-embedding-3-small`; top-10
SMG	Social Memory Graph (JSON-backed graph)	smb310	GPT-4o-mini	Query-type-routed graph traversal; top-10
GRAPHITI	Graphiti-core 0.28.2 + Kuzu 0.11.3	smb310 (Py 3.10)	GPT-4o-mini	FTS + graph; top-10
LANGMEM	LangChain InMemoryStore + reflection	smb311 (Py 3.11)	GPT-4o-mini	Semantic search; top-10
MEM0	Mem0 local + Chroma	smb310	GPT-4o-mini	Semantic; top-10
COGNEE	Cognee batch doc-to-graph	smb310 (Py 3.10)	GPT-4o-mini	Hybrid graph+vector
LLM-MINI	Full context (no memory system)	smb310	n/a	Full corpus
LLM-GEMINI	Full context (no memory system)	smb310	n/a	Full corpus

B.3 Adapter Implementation Notes

SUBJECT-MEM uses the same Chroma vector store as the uncompressed retrieval baseline but changes the unit of storage. At ingestion, GPT-4o-mini parses each session turn and extracts (subject, fact) pairs, where the subject is the person the fact is *about*, not the person who said it. Each pair is stored as "[About: {subject}] {fact}" with speaker metadata. This transforms the indexing from speaker-anchored to subject-anchored without changing the retrieval mechanism, isolating the contribution of ingestion-time attribution.

Uncompressed retrieval baseline. It stores each conversation turn as a single Chroma document with metadata fields `speaker`, `session_index`, and `timestamp`. Retrieval is purely cosine similarity against the question string; no LLM call is made during ingestion. Because raw turn text contains speaker labels verbatim, this baseline scores 0.82 on attribution probes by string matching rather than social reasoning. It is reported as a reference, not as a memory system.

Social Memory Graph (SMG). It maintains a structured JSON graph with node types: `PERSONA` (per-member), `PREFERENCE` (attribute-value facts), `GROUP_NORM` (group-level facts), and `SESSION`. Edge types include: `HAS_PREFERENCE`, `DISSENTS_FROM` (norm exceptions), `KNOWS_ABOUT` (cross-persona knowledge), `EVOLVED_FROM` (superseded preferences), and `PARTICIPATED_IN`. Query-type routing in `eval/adapters/smg/queries.py` directs Q6 queries to `DISSENTS_FROM` traversal and Q5 queries to `KNOWS_ABOUT` traversal, explaining SMG's structural advantage on these types.

Graphiti uses the Graphiti-core library with an embedded Kuzu graph database (no Docker required). During ingestion, each session's turns are submitted as episodes; Graphiti extracts entity-relationship triples using GPT-4o-mini. Two implementation-level bugs were fixed before evaluation: (i) `KuzuDriver._database` was never set, causing an `AttributeError` on `add_episode()`. Fixed by setting `driver._database = network_id` after `KuzuDriver` creation; (ii) Kuzu FTS indexes were not created by the default `build_indices_and_constraints()` in version 0.28.2. Fixed by adding an explicit `CREATE_FTS_INDEX` call for all relevant tables.

LangMem uses LangChain's `InMemoryStore` with a reflection step that consolidates memories after each session. An embedding error in the store's `search()` method silently returned empty results in some configurations. Fixed by adding explicit exception logging and a fallback that returns all memories as flat context.

Mem0 (local) uses the Mem0 Python SDK with local Chroma as the vector backend. An early bug caused per-turn ingestion to strip speaker attribution (storing "User likes X" without the speaker name). Fixed by batch-ingesting whole sessions

using the Mem0 messages API, which preserves the `role`/`content` structure.

C Evaluation Pipeline Details

C.1 CLI and Pipeline

The evaluation pipeline is invoked via:

```
python -m eval.cli --network {network_id} --condition {condition}
```

The pipeline executes four stages: (1) load the conversation and QA files; (2) ingest sessions into the memory system via `adapter.add()`, or assemble full-corpus context for full-context conditions; (3) for each QA pair, call `adapter.search()` and prompt GPT-4o-mini with the retrieved context and question; (4) score each answer against the ground truth.

Results are written to `data/results/{condition}/{network_id}.jsonl` (per-pair) and `data/results/{condition}/{network_id}_summary.json` (summary).

C.2 Answerer Prompts

Separate prompt templates are used for memory-system conditions (context = retrieved memories) and full-context conditions (context = complete corpus). Both templates include query-type-specific answering guidance. For example, the Q8 guidance instructs the model to state the old preference, the new preference, and the trigger. All three are required for full credit.

The memory-system prompt includes explicit instructions for reading SMG-formatted context: "Facts labeled 'session N' are chronological. 'EVOLVED_FROM' / 'Earlier belief (session N)' means the newer fact supersedes it." This ensures the prompt design does not artificially disadvantage structured memory outputs.

C.3 LLM Judge Rubric

Open-ended answers are scored by GPT-4o-mini acting as a judge. The judge receives the question, the gold answer, and the generated answer, and returns a JSON object `{"score": 0.0-1.0, "rationale": "..."}`.

Three critical rules are applied before the general rubric:

1. **Anti-verbosity bias**: a concise correct answer scores the same as a verbose one. Evidence citation is not required for Q1/Q3. This rule was added after observing that early judge versions systematically downscored terse-but-correct responses.
2. **Attribution error**: attributing a fact to the wrong person always scores ≤ 0.3, regardless of how much other content is correct.
3. **Non-answer penalty**: "NOT ENOUGH INFORMATION" or "[No memories retrieved]" always scores 0.0.

Query-type-specific scoring rules are applied on top. Q3 scores are the fraction of group members correctly recalled. Q4 awards 1.0 when the correct speaker is named and the foil is not, 0.7 when the correct speaker is named but the foil is also named, and 0.0–0.3 for a wrong speaker. Q5 requires both the preference and who revealed it through observable chat behavior for a score of 1.0; either part alone scores 0.5. Q8 awards one third of a point each for the old state, the new state, and the trigger; both states without a trigger scores 0.7.

Multiple-choice questions (Q2, Q4, Q6) use exact letter matching (binary 1.0/0.0) with a robust parser that handles formats such as "A", "A.", "Answer: A", and "The answer is A".

C.4 Reproducibility Notes

1. All 43 networks and 1,031 QA pairs are generated by Claude Sonnet 4.5 via Claude Code following the deterministic pipeline in Section 3.3; no API key is required. Each network has an integer seed recorded in `manifest.json`.
2. Model versions: answering model GPT-4o-mini (`gpt-4o-mini-2024-07-18`); full-context baselines GPT-4o-mini and Gemini 2.5 Flash (`gemini-2.5-flash`); judge GPT-4o-mini; SMG and Graphiti extraction GPT-4o-mini. All temperatures are 0.0.
3. Pipeline concurrency: 10 concurrent judge calls, 5 concurrent answer generation calls. Retry policy: exponential backoff, up to 5 retries, maximum delay 60s.
4. Environment: macOS Darwin 25.3.0; Python 3.10 (smb310 env, used for Graphiti and Cognee), Python 3.11 (smb311 env, used for LangMem), Python 3.11 (standard, all other conditions).
5. The dataset is at https://huggingface.co/datasets/anon4data/socialmembench (CC BY 4.0); the evaluation pipeline, generation skills, and

browser-based data audit viewer are at https://anonymous.4open.science/r/SocialMemBench/ (MIT). The repository includes a `requirements.txt` and per-adapter environment specifications. All memory system adapter patches applied before evaluation are included in the repository and described in Appendix B.

6. Exact software versions for evaluated systems: Graphiti-core 0.28.2, Kuzu 0.11.3, Mem0 SDK (local, Chroma backend), LangMem via LangChain `InMemoryStore`, Cognee batch doc-to-graph. Evaluation was conducted April 2026.

D Answerer Model Comparison

GPT-4o-mini is the answerer for all conditions in Table 4. Most deployed memory systems run on smaller models; the matched-answerer choice mirrors that and isolates memory-layer effects from reasoning capacity. Table 10 reports a matched comparison against Gemini 2.5 Flash on the same 10 small-tier networks (77 questions), full-context oracle for both with the same GPT-4o-mini judge. Gemini answers; it does not index or extract.

The aggregate gap is small: 0.075 in Gemini's favour. Per-category gaps are smaller still, under 0.13 on every category except Q3 (MA: +0.30 on $n = 5$, noisy). TM and TS, the categories where memory frameworks score lowest, show gaps of +0.04 and +0.08. GPT-4o-mini reaches TM 0.79 on small-tier; the TM 0.31 in Table 4 reflects the medium- and large-tier rows. The answerer model is not the architectural bottleneck.

Table 11 gives the difficulty-stratified breakdown on the same 10 small-tier networks (77 questions), full-context oracle for both models, with the same GPT-4o-mini judge.

E Pipeline Overview

Figure 4 diagrams the end-to-end SocialMemBench workflow. The left side covers data construction, executed once per network: ego-network generation, conversation generation with implicitly planted social challenges across S sessions, then QA generation with the pre-generation inference gate (Section 3.3), the blind-critic acceptance pass, and human review through the audit viewer. A network only enters the released dataset after all three gates pass. The right side covers evaluation: each condition ingests one session at a time into its memory system, the system's retrieve interface serves a held-out QA pair, and the answerer produces a response that an LLM judge scores against the ground-truth answer. Memory state persists across sessions within a network and is reset between networks, isolating per-network behaviour.

F Data Audit Viewer

The repository includes a browser-based data audit viewer used during dataset construction for human review. It runs locally against the released data files and requires no API key.

```
cd viewer && python server.py
# Open http://localhost:8080
```

The viewer has three tabs. The **Chat** tab renders each network's conversation sessions as a WhatsApp-style thread, with planted-challenge turns highlighted and evidence spans underlined inline; clicking a highlighted turn shows the challenge type, subject persona, and verbatim evidence span. The **Graph** tab shows the persona network as an interactive D3 force graph, with nodes coloured by persona and edges labelled by relationship type, closeness, and sentiment. The **QA** tab lists all QA pairs for the network with their evidence anchors, query type, difficulty label, and QC flags; pairs can be filtered by query type and searched by question text.

During dataset construction, all 1,031 QA pairs were reviewed in this interface to check plausibility, correct evidence attribution, and answer completeness. The viewer is released as part of the benchmark so that reviewers and future users can inspect any network and verify ground-truth anchors without writing code.

G Representative Prompts

The following prompts are reproduced verbatim from `eval/config/prompts.yaml` and `eval/adapters/subject_rag_adapter.py`. Template variables (`{context}`, `{question}`, etc.) are filled at runtime.

G.1 Open-Ended Answerer (Memory System Conditions)

This prompt is used when a memory system's retrieved output is passed to GPT-4o-mini as context. It instructs the model to use only what was retrieved and provides per-category answering guidance.

```
You are evaluating a social group AI memory system.
You have retrieved memories from a group chat and must answer
an open-ended question. Answer ONLY using the retrieved
memories. Use no outside knowledge or guesses.
```

Table 10: Answerer-model comparison: per-category mean scores on 10 small-tier networks (77 questions), full-context oracle for both models, same GPT-4o-mini judge. DM (Q9) is omitted because the 10-network subset does not contain member-departure scenarios; DM scores are reported in Table 4.

Model	**Mean** (n=77)	**SR** (15)	**GD** (5)	**MA** (5)	**AP** (8)	**TM** (10)	**NI** (8)	**RE** (10)	**TS** (16)
GPT-4o-mini (full ctx)	0.644	0.67	0.60	0.34	0.50	0.79	1.00	0.65	0.53
Gemini 2.5 Flash (full ctx)	0.719	0.76	0.60	0.64	0.62	0.83	1.00	0.68	0.61

Table 11: Difficulty-stratified breakdown of the answerer-model comparison. Difficulty levels follow the inference-depth rubric (Section 3.3): easy = single-turn lookup, medium = 2-turn synthesis or one inference step, hard = cross-session inference or attribution ambiguity.

Model	**Mean**	**Easy** ($n = 6$)	**Medium** ($n = 57$)	**Hard** ($n = 14$)
GPT-4o-mini (full ctx)	0.644	0.95	0.66	0.44
Gemini 2.5 Flash (full ctx)	0.719	0.97	0.73	0.58

Table 12: Question-weighted mean score by network-size tier. Tier definitions and question counts are given in Table 3; a visualisation of the U-shape across tiers is in Figure 2.

Condition	**Small**	**Medium**	**Large**
Reference conditions			
LLM-MINI	0.476	0.257	0.395
UNCOMPRESSED RET.	0.369	0.298	0.369
Memory frameworks			
GRAPHITI	0.215	0.159	0.170
LANGMEM	0.194	0.147	0.154
MEM0	0.156	0.125	0.150
COGNEE	0.123	0.138	0.104
Research probes			
SUBJECT-MEM	0.313	0.276	0.362
SMG	0.249	0.179	0.219

```
Pay attention to WHO said what and WHEN.
Attribution and timing matter.

Answering guidance by question type:
- Single preference (Q1): Name the person and state their
  preference. Be specific.
- Group decision (Q2): State the decision AND name any
  dissenter(s); implicit dissent counts.
- Multi-contact (Q3): Cover every group member; say "unclear"
  for those with no signal.
- Theory of mind (Q5): State BOTH the preference AND who
  revealed it, with the specific observable action that shows
  they already knew.
- Temporal shift (Q8): State the OLD preference, the NEW
  preference, and what triggered the change.

[MEMORIES]
{context}

[QUESTION]
{question}
```

G.2 LLM Judge

The judge receives the question, the gold answer, and the generated answer and returns a JSON score. Three critical rules apply before the general rubric: brevity is not penalised, wrong attribution caps the score at 0.3, and a refusal to answer scores 0.0.

```
Score the generated answer against the gold answer
on a 0.0-1.0 scale.

CRITICAL RULES:
1. A concise correct answer scores THE SAME as a verbose one.
2. Attributing a fact to the wrong person -> score <= 0.3.
3. "NOT ENOUGH INFORMATION" always scores 0.0.

General rubric:
- 1.0: Core fact correct, correct attribution, all parts present
- 0.7-0.9: Right answer but one minor part missing
- 0.4-0.6: Right direction but wrong detail or missing element
- 0.0-0.3: Wrong attribution, wrong fact, or no answer

Q-type rules:
- Q3: score = fraction of group members correctly recalled
- Q4: correct speaker + foil not named → 1.0; foil also
  named → 0.7; wrong speaker → 0.0-0.3
- Q5: both preference AND observable evidence of prior
  knowledge required for 1.0; one part → 0.5
- Q8: old state (+0.33) + new state (+0.33) +
  trigger (+0.33) = 1.0

Question: {question}
Gold answer: {golden_answer}
Generated answer: {generated_answer}

Output JSON only:
{"score": 0.0, "rationale": "one sentence"}
```

G.3 Subject-Indexed Fact Extraction (Subject-Mem)

At ingestion time, Subject-Mem calls GPT-4o-mini once per turn to extract (subject, fact) pairs, where

the subject is the person a fact is *about*, not the person who said it. This single change to the indexing step is the only difference from the uncompressed retrieval baseline.

```
You are reading a message from a group chat. Extract any
facts about specific people mentioned.

Message: [{speaker}]: {message}

For each fact, output a JSON object:
  "subject": the person the fact is about
  "fact": a concise statement of the fact

Return a JSON array. Return [] if no person-specific facts.
Only include facts clearly about a named person.

Examples:
Input: [Nadia]: Rafe always goes to the pub after long runs
Output: [{"subject": "Rafe",
          "fact": "always goes to the pub after long runs"}]

Input: [Tom]: Great session today everyone
Output: []
```

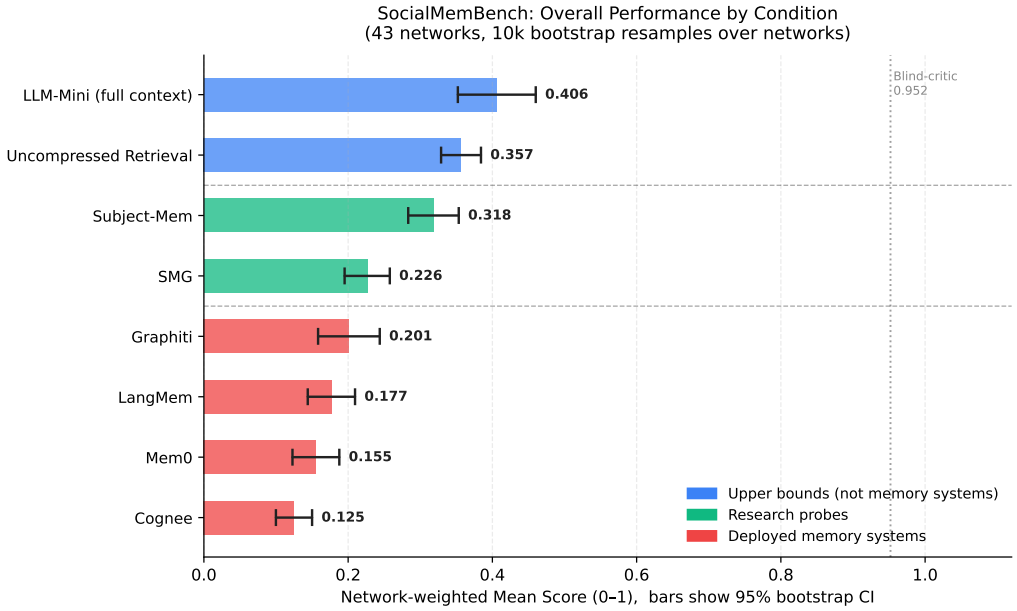


Figure 1: Network-weighted mean score by condition across all 43 evaluated networks, with bootstrap 95% confidence intervals. All four memory frameworks score below the uncompressed retrieval reference, with overlapping CIs between adjacent frameworks.

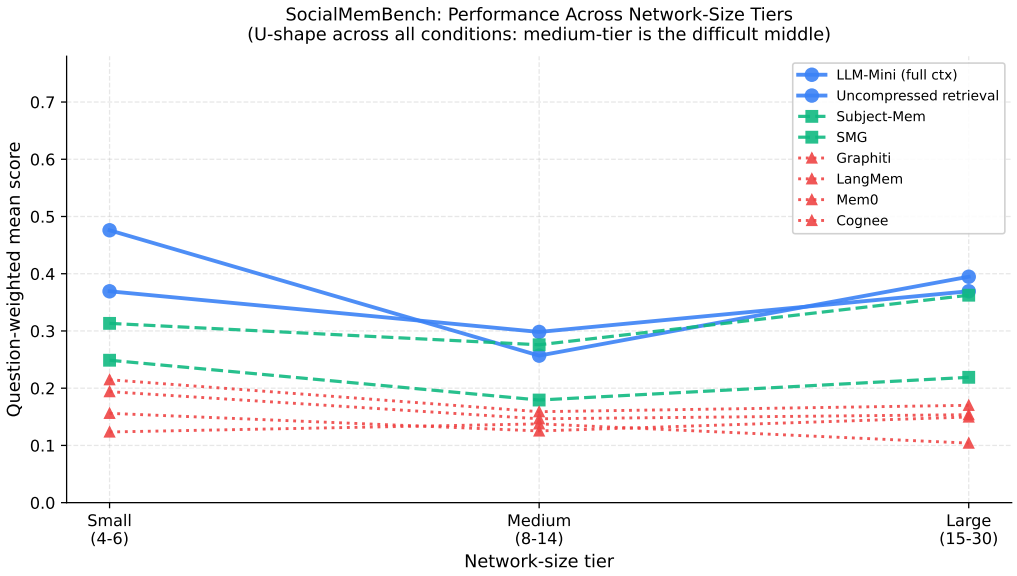


Figure 2: Performance across network-size tiers (visualisation of Table 12). Most conditions follow a U-shape with medium-tier as the difficult middle: small-tier corpora concentrate signal per persona, large-tier corpora compensate with more sessions and turns providing supporting attribution evidence. Memory frameworks plateau at 0.10–0.17 across medium and large; the uncompressed retrieval baseline and research probes recover meaningfully at large scale. Cognee is the only condition that declines monotonically with scale.

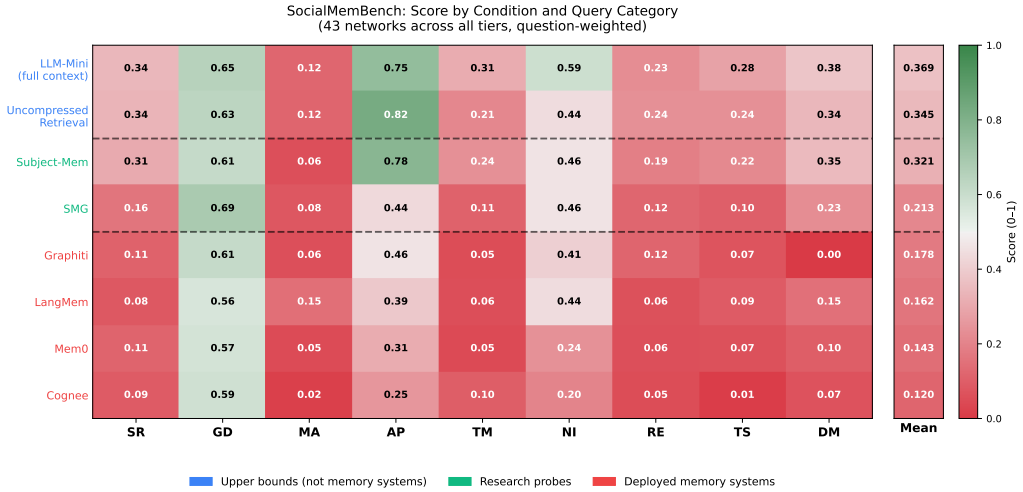


Figure 3: Score by condition and query category (43 networks, question-weighted). Rows are ordered from full-context references (blue) through research probes (green) to memory frameworks (red). The rightmost axis shows each condition's mean score. SMG leads on group decisions (GD: 0.69) and subject-mem leads on attribution probes (AP: 0.78); TM and TS scores are near floor across all four memory frameworks; on DM (departed-member) Graphiti scores 0.00 while Subject-Mem retains 0.35.

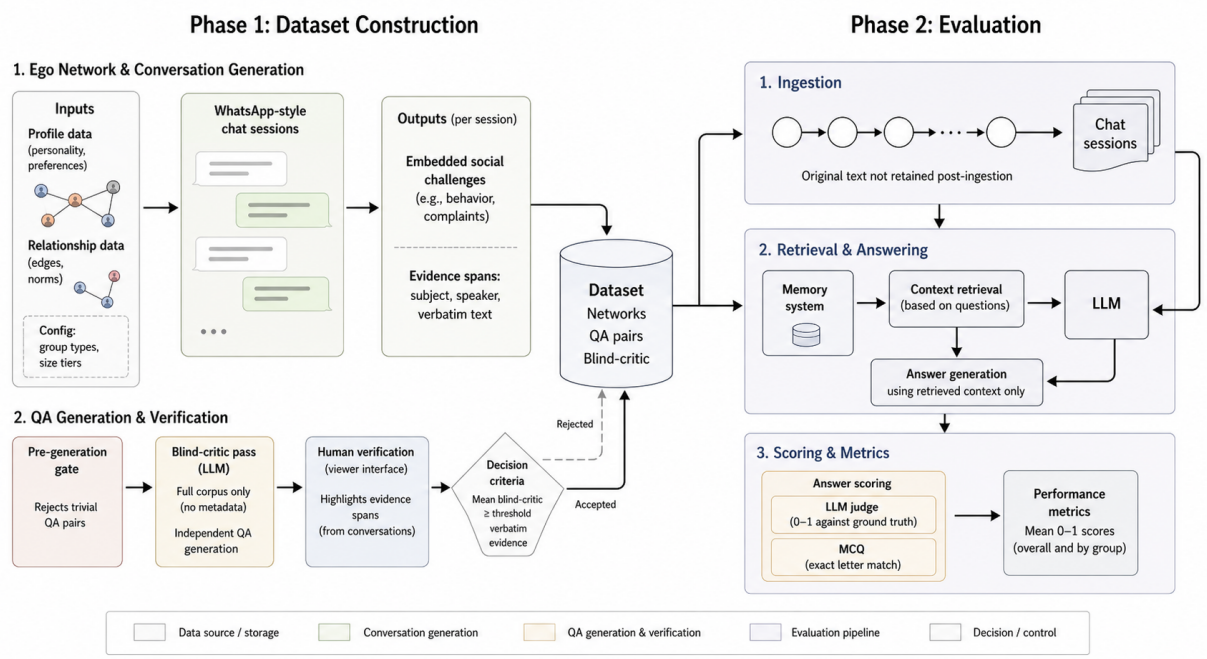


Figure 4: SocialMemBench pipeline. Phase 1 (left) covers dataset construction: persona network and conversation generation with implicitly planted social challenges, followed by QA generation with a pre-generation gate, blind-critic pass, and human verification via the data audit viewer. Phase 2 (right) covers evaluation: session-by-session ingestion into a memory system, retrieval and answering using only retained context, and LLM judge scoring.

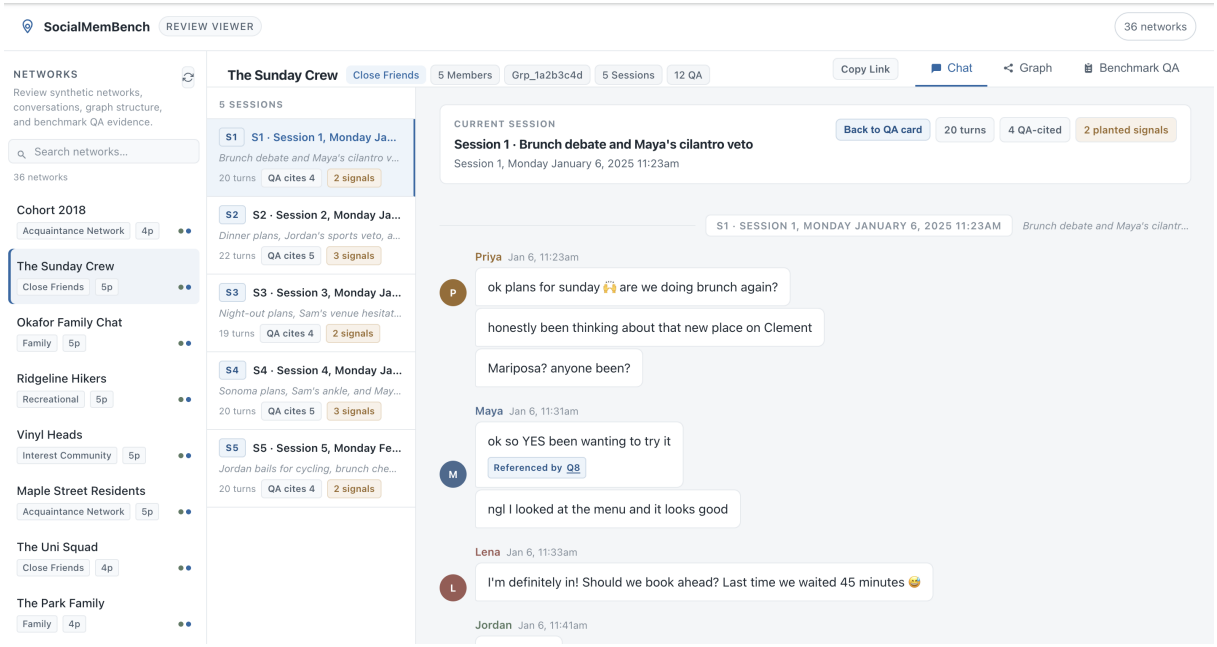


Figure 5: Data audit viewer showing the Chat tab. Turns containing planted challenges are highlighted; clicking a highlighted turn opens the challenge panel (right) with the challenge type, subject persona, and verbatim evidence span. The QA tab (not shown) links each question to its evidence anchors in the same interface.